

Multi-Source Candidate Data Transformer — Technical Design

Eightfold Engineering Intern Assignment (Jul–Dec 2026) | Shalu Kumari | shalujha6044@gmail.com

1. Pipeline Flow

detect → **extract** → **normalize** → **match/cluster** → **merge+confidence** → **project** → **validate** → **emit**

- **detect**: Inspects file headers and extensions to route to source-specific loaders (errors are logged and skipped, never fatal).
- **extract**: Parses raw data into standardized fields tagged with metadata (source_id, source_type, method).
- **normalize**: Independently normalizes individual fields (phones, dates, countries, skills) to canonical shapes.
- **match/cluster**: Groups related records into disjoint candidate-clusters using a *Union-Find* algorithm.
- **merge+confidence**: Resolves conflicts using source-trust and extraction-method weights, scoring field-level confidence.
- **project & validate**: Evaluates runtime config paths, executes mapping/normalizations, enforces schema constraints, and emits JSON.

2. Canonical Schema & Normalized Formats

The internal representation of a candidate profile consists of standard structured fields:

- **phones**: Standardized to **E.164** via the *phonenumbers* library. If missing country codes, falls back to a default region. Ambiguous cases resolve to null with a provenance note (never guessed).
- **dates**: Parsed to **YYYY-MM**. If year-only dates are ingested (e.g. '2020'), they are normalized to '2020-01' with a *precision: year* provenance flag to avoid presenting fabricated information as fact.
- **country**: Normalized to **ISO-3166 alpha-2** via standard country name and alias lookup tables.
- **skills**: Normalized via a canonical dictionary (e.g., 'ReactJS' → 'react'). Unrecognized skills are lowercased and preserved with lower confidence.

3. Matching, Merge Policy & Confidence

Disjoint candidate records are clustered prior to field merging to prevent cross-contamination:

- **Clustering Keys**: Match on normalized emails → normalized phones → conservative fuzzy name + current_company. Near-duplicates (same name and company, but differing emails/phones) are explicitly prevented from auto-merging.
- **Confidence Model**: Field confidence is defined as $source_trust \times method_confidence$. Source trust values favor objective inputs: *recruiter_csv* (0.90) and *ats_json* (0.85) over *resume* (0.75) and *recruiter_notes* (0.55). Method confidence values are: *direct_field* (1.0), *regex_extract* (0.8), *heuristic_parse* (0.6), and *derived* (0.9).
- **Corroboration & Conflict**: Agreement on the same normalized value by multiple independent sources triggers a corroboration boost (+0.05 per source, capped at 1.0). The winning value is emitted, while all losing values are fully retained in the provenance log to prevent silent data loss.

4. Runtime Output Config

A separate projection layer consumes the immutable CanonicalRecord through a config JSON to shape output: it selects field subsets, remaps paths (using dot/bracket notation like *emails[0]* or *skills[.name]*), implements local normalization overrides, and toggles provenance and confidence metadata. The projected result is validated against the config-specified types. Path configurations are validated against the canonical schema at load time to fail fast rather than failing silently at runtime.

5. Edge Cases Handled

- **Near-Duplicates**: Candidates with same name and company but different emails remain separate profiles, preventing incorrect identity merge.
 - **Stated vs. Computed Experience**: Date math over job intervals produces a computed value (method: *derived*). This wins over stated experience (method: *regex_extract*) due to its higher method weight, with conflicts preserved.
 - **Ambiguous Phone Numbers**: Phone numbers lacking country codes that cannot be resolved default to null with an explanatory note.
 - **Garbage Resume File**: Unreadable files fail closed; the extractor logs the error, contributes zero fields, and other sources proceed.
 - **Config Path Typo**: Configurations referencing invalid fields are rejected immediately during load time.

6. Deliberately Out of Scope

- **Name-Only Matching**: Excluded due to high false-positive risks which could corrupt critical hiring decisions.
 - **Live API Integrations**: GitHub/LinkedIn APIs are simulated via structured and unstructured input adapters to fit the offline sandbox.
 - **User Interface**: Focus is kept entirely on core pipeline correctness, regression testing, and the CLI surface.